

Maximize the Minimum Powered City

There are n cities, numbered from 0 to $n - 1$, each containing some power stations. In particular, the i -th city contains `stationsi` power stations.

All power stations have a range r . A power station in the i -th city provides power to the j -th city if $|i - j| \leq r$. The *power* of a city is the number of power stations it receives power from.

You are told to build k additional power stations, each with the same range r . You may build these extra stations in any city, and you can build multiple stations in the same city. Once you have added the extra stations, consider the resulting arrangement of all power stations. The *strength* of this arrangement is the minimum power across all n cities.

You are given n , the array `stations`, as well as the integers r and k . Compute the maximum strength across all possible resulting arrangements.

Example Test Cases:

Test Case 1

Input: $n = 5$, `stations` = `[1,2,4,5,0]`, $r = 1$, $k = 2$

Output: 5

Explanation: Install 2 additional stations in city 1. The station counts for each city become `[1,4,4,5,0]`. The powers of the cities in the resulting arrangement become `[5,9,13,9,5]`. The strength of the arrangement is 5.

Test Case 2

Input: $n = 4$, `stations` = `[4,4,4,4]`, $r = 0$, $k = 3$

Output: 4

Explanation: Each power station can only power its own city. Then, regardless of where the additional stations are built, a city with a power of 4 will always remain.

Expected Time Complexity: $O(n \log(\Sigma + k))$

(where Σ is the total number of power stations initially)

Input Format:

An integer n , an array `stations` of length n , and two integers r and k .

Output Format:

Return a single integer, the maximum possible strength, as defined above.

Constraints:

1. $1 \leq n \leq 10^5$

2. $0 \leq \text{stations}[i] \leq 10^5$

3. $0 \leq r \leq n - 1$

4. $0 \leq k \leq 10^9$

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the maximum strength.

The `solve` function must compute this in $O(n \log(\Sigma + k))$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either `C++` or `python`. For `C++`, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.